

My Product Journey: Milk — Product Fact Card

How to use this card: This document provides detailed information about milk in India. Read it carefully to extract facts for your graphic organiser.

Why Milk Matters — and Where It Comes From

Milk stands as a fundamental everyday product in India, which proudly holds the title of the world's largest milk producer, contributing approximately 24% of the global output. This vital dairy sector significantly underpins the Indian economy, contributing about 5% to the national GDP and providing a crucial livelihood for over 80 million farmers, a majority of whom are women. Most of the milk produced, with a per capita availability of 471 grams per day in 2023-24, is consumed domestically. The historical 'White Revolution,' also known as Operation Flood, spearheaded by Dr. Verghese Kurien starting in 1970, was instrumental in transforming India into a self-sufficient and eventually the world's leading milk producer by establishing a robust cooperative dairy network. The heartland of milk production spans across states like Uttar Pradesh, Rajasthan, Andhra Pradesh, Gujarat, and Punjab, which are the top five contributors. Key organisations like Amul (Gujarat Cooperative Milk Marketing Federation - GCMMF), Mother Dairy, Nandini (Karnataka Cooperative Milk Producers' Federation - KMF), Aavin (Tamil Nadu Cooperative Milk Producers' Federation), Parag Dairy (Uttar Pradesh), Saras Dairy (Rajasthan Cooperative Dairy Federation - RCDF), the National Dairy Development Board (NDDB), and the Food Safety and Standards Authority of India (FSSAI) play pivotal roles in this extensive industry.

The Journey: From Raw Milk to Packaged Milk

The journey of milk begins with **Dairy Farming/Production** in rural areas across India, such as Meerut, Muzaffarnagar, Bulandshahr, and Aligarh in Uttar Pradesh, or East Godavari, Chittoor, and Guntur in Andhra Pradesh. Here, dairy farmers, often smallholder or marginal, along with farmhands and cattle rearers, milk cows and buffaloes either manually using milking pails or with milking machines. Veterinary assistants ensure animal health, while farmers use chaff cutters and fodder choppers for feed preparation. The raw milk, valued at a farm-gate price of Rs 36.7-49.2 per litre depending on the type, then moves to **Collection** centers. At village-level Milk Collection Centers (MCCs) or Dairy Cooperative Societies (DCS), milk collection agents receive the milk. Society secretaries oversee operations, while milk testers use weighing scales, lactometers, and fat testing machines like the Milko-tester to assess fat and Solids-Not-Fat (SNF) content, which determines farmer payment. Data entry operators record these details. This stage adds approximately Rs 1-2 per litre for initial handling and administration. Next, for **Chilling & Transportation**, chilling plant operators at Bulk Milk Coolers (BMCs) rapidly cool the raw milk to 4°C or lower to inhibit bacterial growth. Tanker drivers and their helpers then collect this chilled milk in insulated stainless steel milk tankers, which can hold 2,000 to 40,000 litres, for transport to larger processing plants. Logistics coordinators manage these routes. This stage incurs costs of Rs 3-5 per litre for chilling, fuel, and vehicle maintenance. Upon arrival at **Processing** plants in locations like Anand, Gujarat, Delhi-NCR, Mumbai, Bengaluru, Chennai, or Lucknow, dairy technologists and plant operators oversee reception, filtration, and clarification. Quality

control chemists ensure standards are met. The milk is then standardized by plant operators using cream separators to adjust fat and SNF content, followed by pasteurization using Plate Heat Exchangers (PHE) at 72°C for 15 seconds (HTST) or 135-150°C for 2-4 seconds (UHT). Homogenization, performed by plant operators using homogenizers, breaks down fat globules. Finally, packaging machine operators direct the milk into pouch fillers or Tetra Pak machines. Maintenance staff and sanitation workers ensure plant upkeep. This processing adds Rs 5-8 per litre. The final stage is ****Distribution & Retail****, where warehouse managers receive packaged milk and delivery drivers transport it in refrigerated vans to regional warehouses and then to various retail outlets such as kirana stores, supermarkets, hypermarkets, and online platforms. Retail staff and merchandisers stock the milk in retail display chillers. This stage adds Rs 8-12 per litre for logistics, marketing, and retail margins.

Raw Materials, Inputs, and Natural Resources

The primary raw materials for India's milk industry are raw cow milk and buffalo milk. Cow milk, contributing approximately 43% of the total milk production, is sourced across India, predominantly from states like Uttar Pradesh, Rajasthan, Madhya Pradesh, Punjab, Haryana, Gujarat, Maharashtra, Andhra Pradesh, Karnataka, Tamil Nadu, and Bihar. Indigenous breeds like Gir, Rath, Tharparkar, and Kankrej, alongside crossbreds such as Holstein Friesian and Jersey, are common. The average farm-gate price for cow milk is projected at Rs 36.7 per litre for 2025-26. Buffalo milk, which accounts for a larger share at 48%, is mainly produced in Uttar Pradesh, Punjab, Haryana, Gujarat, and Andhra Pradesh, with Murrah buffaloes being a high-yielding breed known for higher fat content. Its average farm-gate price is estimated at Rs 49.2 per litre for 2025-26. Goat milk contributes a smaller but regionally significant 4%, primarily from arid and semi-arid regions. Beyond the milk itself, cattle feed and fodder represent a major input cost for farmers, including green fodder, dry fodder, and concentrate feed, with rising costs and scarcity being significant concerns due to shrinking grazing lands and erratic monsoons. The production of milk also has a substantial water footprint, estimated at approximately 1078 litres per kilogram of milk, with 90-95% of this attributed to feed and fodder cultivation. Direct water use in milk processing facilities is around 3.31 litres per kilogram of milk processed, with an average total water footprint, including indirect use for electricity, reaching 9.0 litres per kilogram. Groundwater over-extraction for agriculture, including fodder, remains a major environmental concern, particularly in North-Western, Western, and Southern India.

How Milk Is Made — Step by Step

The transformation of raw milk into safe, packaged milk involves several precise steps within a processing plant:

- **Reception & Testing:**** Upon arrival at the processing plant, dairy technologists and quality control chemists receive the raw milk from tanker drivers. They meticulously test the milk for various quality parameters, including fat and Solids-Not-Fat (SNF) content, and check for adulterants before it is accepted for processing.
- **Filtration & Clarification:**** Plant operators then direct the raw milk through filtration units to remove any visible impurities. Following this, clarification centrifuges are used by plant operators to further remove suspended solids, ensuring a cleaner product.
- **Standardization:**** Dairy technologists and plant operators utilize cream separators to adjust the milk's fat and SNF content to meet specific product requirements, such as full cream, toned, or double-toned milk.
- **Pasteurization:**** Plant operators manage Plate Heat Exchangers (PHE) to heat the milk. For High-Temperature Short-Time (HTST) pasteurization, the milk is heated to 72°C and held for 15 seconds. Alternatively, for Ultra-High Temperature (UHT) processing, it is heated to a much higher 135-150°C for 2-4 seconds, effectively eliminating harmful bacteria while preserving nutritional value.
- **Homogenization:**** After pasteurization, plant operators guide the milk through homogenizers. These

machines apply high pressure to break down the fat globules into much smaller, uniform sizes, preventing the cream from separating and rising to the top. 6. **Cooling & Storage:** Following homogenization, plant operators rapidly cool the milk to a low temperature, typically 4°C, to maintain its freshness and inhibit bacterial growth. The cooled milk is then temporarily stored in large, sterile storage tanks. 7. **Packaging:** Finally, filling machine operators and packaging machine operators direct the processed milk into the designated packaging, such as plastic pouches using pouch fillers or aseptic cartons using Tetra Pak machines, sealing it for distribution and sale.

What It Costs to Bring Milk to You

The final retail price of milk in India is a cumulative sum built up through various stages of its journey. The process begins with the farmer's share, which constitutes the largest portion. A dairy farmer receives approximately Rs 36.7 per litre for cow milk and Rs 49.2 per litre for buffalo milk (based on 2025-26 average farm-gate prices), representing about 75% of the final retail price. As the milk moves to collection centers, an additional Rs 1-2 per litre is added for initial handling, testing, and administrative overhead by milk collection agents and society secretaries. The subsequent stage of chilling and transportation, managed by chilling plant operators and tanker drivers, adds another Rs 3-5 per litre to cover costs like energy for cooling, fuel for insulated tankers, and vehicle maintenance. At the processing plants, dairy technologists and plant operators contribute Rs 5-8 per litre for energy, labor, chemicals, equipment depreciation, and quality control measures during pasteurization, homogenization, and packaging. Finally, the logistics and distribution, handled by warehouse managers, delivery drivers, and sales representatives, add Rs 3-5 per litre for transport in refrigerated vans and cold chain maintenance. The retailer margin and marketing efforts by retail staff and merchandisers contribute a further Rs 2-4 per litre, bringing the total retail price for various types of pouch milk to a range of Rs 54-80 per litre (as of May 2026). Regarding taxation, fresh, unbranded, unflavored, raw, pasteurized, and UHT milk are exempt from Goods and Services Tax (GST) at 0%. However, flavored milk, condensed milk, and milk powder are typically taxed at 5% or 12%, while other dairy products like butter, cheese, and ghee attract higher GST rates ranging from 5% to 18%.

What Milk Costs in Shops — Price Research

The cost of milk can vary significantly across different brands, types, and retail channels in India, reflecting regional pricing strategies and market dynamics. For example, on May 14, 2026, a consumer in Delhi-NCR or Uttar Pradesh would pay approximately Rs 72.0 for a 1-litre pouch of Amul Gold Full Cream Milk at a local kirana store or supermarket. In contrast, Mother Dairy Toned Milk, also a 1-litre pouch, was priced at Rs 60.0 at a local kirana store or Mother Dairy Booth in Delhi-NCR on the same date. The difference here often reflects the fat content, with 'Full Cream' milk typically being more expensive than 'Toned' milk, and the efficiency of a brand's dedicated retail outlets like Mother Dairy booths. Moving south, the pricing structure changes due to the strong presence of state cooperative federations. In Karnataka, a 1-litre pouch of Nandini Toned Milk (Blue Packet) was available for Rs 46.0 at a local kirana store or supermarket on March 27, 2025. Even more notably, in Chennai, Tamil Nadu, a 1-litre pouch of Aavin Toned Milk (for non-card holders) was priced at a significantly lower Rs 20.5 at an Aavin Depot or retailer on September 1, 2025. The substantial price difference for Nandini and Aavin milk compared to national brands like Amul and Mother Dairy is primarily due to these being state cooperative brands that often receive government subsidies, allowing them to offer milk at more affordable rates to support local farmers and consumers. These price variations highlight the impact of brand positioning, fat content, distribution models, and regional cooperative support on the final retail price of milk.

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